

A
Project Report
ON
Calories Burnt Prediction Web-Application
Submitted The Partial Fulfillment of The Award Of
Master Of Computer Application

Submitted By:

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(2024-2025)

Declaration

I hereby declare that the project work entitled “**Calories Burnt Prediction Web-Application (Machine learning model)**” is an authenticated work carried out by me at Government College for Girls, Sector-14, Gurugram and this work has not been submitted for similar purpose anywhere else.

Date: 29/04/2025

Class: MCA-2nd year

Student Name: Sandhya Chauhan

Roll No: 2231914032

Certificate

This is to certify that **Sandhya Chauhan** roll no.**2231914032** has worked under my supervision to prepare her project report. The work embodied in this report is original and was conducted at Government College for Girls, Gurugram. She has completed all requirements of MCA ordinance.

(Project Guide)

Mrs. Nirmal Godara

Assistant Professor, Computer Science

GCG, sec-14, Gurugram

Acknowledgement

I would like to express my special thanks of gratitude to my teacher and project guide **Mrs. Nirmal Godara** for her valuable guidance. I am also thankful all the faculty member of the Computer Science Department for their encouragement and help while collecting information about the project.

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Problem Statement

Overview:

The aim of this project is to design and develop a web-based application that accurately predicts the number of calories burnt during various physical activities, based on user inputs. The application will allow users to input personal details (such as age, gender, weight, and height) and activity data (such as type of exercise, duration, and intensity), and receive an estimate of the calories burned. This prediction will help users make informed decisions about their fitness routines, track their progress, and manage their health and wellness goals.

Necessity of the Web Application:

1. **Health and Fitness Awareness:** In today's world, maintaining a healthy lifestyle is a growing concern. People are increasingly seeking ways to track their health and fitness progress. One crucial aspect of fitness is understanding how much energy (calories) they burn during exercise. This knowledge is essential for managing weight, improving athletic performance, and achieving health goals (e.g., weight loss, muscle gain, improving cardiovascular health).
2. **Personalized Fitness Recommendations:** Different people have varying fitness goals and needs. A generalized estimate of calories burnt may not be helpful. Personalization is key. By providing users with a tool that considers their age, gender, weight, and activity intensity, the application can give a more accurate estimate of calories burned, making it a valuable tool for individuals who are serious about their fitness journey.
3. **Informed Decision-Making:** Many people engage in physical activities without a clear understanding of how much energy they are expending, which can lead to under or overestimating the effort required to achieve fitness goals. Having a reliable calorie burn estimate allows users to make better decisions on how to structure their workouts, whether they want to increase intensity, adjust duration, or focus on specific goals like fat loss or muscle gain.
4. **Track Progress:** Tracking calories burnt is a key component of long-term fitness tracking. Users want to see improvements, whether in endurance, strength, or overall health. This web application can help users monitor their activity levels over time, giving them insights into their fitness progress.
5. **Accessibility:** Traditional methods for tracking calories burnt require sophisticated equipment (e.g., heart rate monitors, wearables) or advanced

knowledge of exercise physiology. This web application will democratize access to calorie burn estimates by providing a simple, user-friendly platform that is accessible to anyone with an internet connection.

6. **Health and Weight Management:** A significant portion of people engage in physical activities to control their weight, improve their metabolism, or achieve other health-related goals. Accurate calorie burn predictions can help individuals better understand their energy balance (calories consumed vs. calories burnt) and make more informed decisions about diet and exercise. This will lead to more effective weight management and improved overall health outcomes.

Key Features:

1. **User Inputs:**
 - **Personal Details:** Age, gender, weight, height.
 - **Activity Details:** Type of exercise (e.g., running, cycling, swimming), duration, and intensity level (low, moderate, high).
2. **Activity Database:** A comprehensive database of various physical activities and their corresponding calorie burn rates at different intensity levels.
3. **Prediction Algorithm:** An accurate algorithm or machine learning model that takes user data to calculate an estimate of calories burnt. The algorithm could utilize metabolic rate, intensity factors, and exercise duration to predict energy expenditure.
4. **Real-Time Results:** Immediate feedback on the calories burnt after the user submits their data.
5. **Progress Tracking:** A feature to track the user's activity history and visualize trends over time (e.g., calories burnt per week/month).
6. **Recommendations:** Based on the user's fitness goals (e.g., weight loss, muscle gain), the app can suggest activities or adjustments to workouts to maximize results.
7. **Data Privacy and Security:** Protection of user data with secure logins and encrypted data storage, ensuring privacy and confidentiality.

Introduction

1.1 Scope

1.2 Definitions, Acronyms and Abbreviations

1.3 Overview

1.1 Scope

The **Calories Burnt Prediction** project, when applied in the real world, can have far-reaching implications in the health, fitness, and wellness sectors.

- **Personal Fitness Tracking:** Users can track their daily activities and estimate calories burned during various physical exercises (e.g., running, cycling, swimming).
- **Health Monitoring:** Helps individuals monitor their energy expenditure, contributing to better weight management, fitness levels, and overall health.
- **Sports & Fitness Industry:** Fitness enthusiasts, athletes, and trainers can use the app to fine-tune workout routines, track performance, and manage energy balance for optimal results.
- **Corporate Wellness Programs:** Employers can use the tool to promote employee health, encourage physical activity, and help workers stay active as part of corporate wellness initiatives.
- **Diet & Weight Management:** Integrating with diet tracking apps or plans, users can balance calorie intake with calories burnt, supporting weight loss, maintenance, or muscle gain goals.
- **Health Education:** Informs individuals about the energy costs of different activities, empowering them to make healthier lifestyle choices.
Behavioral Change: Encourages more physical activity by making users aware of the caloric benefits of exercise, motivating them to be more active.
- **Scientific Research:** Can be used as a tool in health and fitness studies to better understand energy expenditure and its relationship with fitness levels across different populations.

Future Scope

- **Integration with Health Apps:** Integration with other health apps for a holistic view of fitness and nutrition.
- **Advanced Machine Learning Models:** Using AI and machine learning to predict more personalized calorie burn patterns and offer smarter workout recommendations.

1.2 Definitions, Acronyms and Abbreviation

BMI (Body Mass Index):

- **Definition:** A number calculated from a person's height and weight, used to assess whether they are underweight, normal weight, overweight, or obese.

API (Application Programming Interface):

- **Definition:** A set of tools and protocols that allow different software applications to communicate with each other.

UI (User Interface):

- **Definition:** The visual and interactive elements of a web application that allow users to interact with the system.

UX (User Experience):

- **Definition:** The overall experience of a user when interacting with a product, particularly with regard to how easy or enjoyable it is to use.

HTTPS (Hyper Text Transfer Protocol Secure):

- **Definition:** A protocol used for secure communication over a computer network, especially on the internet.

JSON (JavaScript Object Notation):

- **Definition:** A lightweight data format that is easy for humans to read and write, and easy for machines to parse and generate.

SSL (Secure Sockets Layer):

- **Definition:** A protocol for establishing secure communication between a web server and a browser.

HTTP (Hyper Text Transfer Protocol):

- **Definition:** The protocol used for transferring web pages over the internet.
- **Usage in Project:** HTTP will be used for communication between the client (user's browser) and the server to retrieve and submit data.

ML (Machine Learning):

- **Definition:** A subset of artificial intelligence (AI) that enables systems to learn and improve from experience without being explicitly programmed.

SaaS (Software as a Service):

- **Definition:** A software distribution model where applications are hosted by a service provider and accessed over the internet.

1.3 Overview

The **Calories Burnt Prediction Web Application** is designed to help users estimate the number of calories they burn during physical activities. This tool is particularly beneficial for individuals who want to track their fitness progress, manage their weight, or optimize their workout routines. By providing accurate, personalized calorie burn estimates based on user input (such as weight, age, gender, and activity type), the app helps users make informed decisions about their exercise routines and overall fitness goals.

Advantages:

1. Personalized Fitness Insights:
2. Simple and Intuitive Interface:
3. Helps with Goal Setting:
4. Instant Feedback:
5. Progress Tracking:
6. Promotes Health Awareness:
7. Accessible Anytime, anywhere:

Disadvantages of the Project:

1. Limited Accuracy of Prediction:
2. Dependency on Self-Reported Data:
3. Does Not Account for External Factors
4. Lack of Wearable Integration:
5. No Diet or Nutrition Tracking

Software Requirement Specification

- **Data Flow Diagram**
- **Use Case Description**

SRS (Software Requirements Specification) is a detailed document that describes the **functional and non-functional requirements** of a software project. It outlines what the system will do and how it is expected to perform.

Now, applying this to a **Calories Burnt Prediction** system:

SRS for Calories Burnt Prediction Project

1. Introduction

1.1 Purpose

The purpose of this project is to develop a machine learning-based system that predicts the number of calories burnt by an individual based on personal and activity-related parameters.

1.2 Scope

The system will collect user inputs such as age, gender, height, weight, duration of activity, and type of activity. Based on these inputs, the model will predict calories burnt. The software can be used in fitness apps, health monitoring systems, or wearables.

1.3 Intended Audience

- Software developers
- Data scientists
- Project managers
- End users (fitness enthusiasts, health professionals)

2. Overall Description

2.1 Product Perspective

This is a standalone application or a module that can be integrated with health or fitness platforms.

2.2 Product Functions

- Accept input data from users
- Preprocess data
- Use trained machine learning model to predict calories burnt
- Display predicted output

2.3 User Characteristics

Basic familiarity with health metrics (age, weight, etc.), no technical expertise required.

2.4 Assumptions and Dependencies

- A dataset for training the model is available (e.g., GitHub Calories Burnt dataset)
- Internet connection may be required if cloud-based
- Device must support necessary libraries/frameworks (e.g., Python, Flask, TensorFlow)

3. Specific Requirements

3.1 Functional Requirements

- **FR1:** The system shall accept input data: age, gender, height, weight, activity type, and duration.
- **FR2:** The system shall validate inputs for correctness (e.g., age between 10 and 100).
- **FR3:** The system shall preprocess inputs before feeding into the ML model.
- **FR4:** The system shall return the predicted calorie expenditure.
- **FR5:** The system shall allow users to view historical predictions (optional).

3.2 Non-Functional Requirements

- **NFR1:** The UI shall be responsive and mobile-friendly.

- **NFR2:** Data privacy must be maintained (comply with GDPR if in EU).
- **NFR3:** The prediction model must have an accuracy of at least 85% on test data.

4. System Design Requirements

4.1 Hardware Interface

- PC or smartphone with at least 2GB RAM
- Internet browser (for web version)

4.2 Software Interface

- Python 3.x
- Flask/Django (if web-based)/Fast API
- Scikit-learn / TensorFlow / PyTorch
- HTML/CSS/JavaScript (for frontend)

5. Future Enhancements

- Integration with wearable devices (e.g., Fitbit, Apple Watch)
- Real-time calorie burns tracking
- Personalized fitness recommendations

- **Data Flow Diagram:**

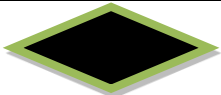
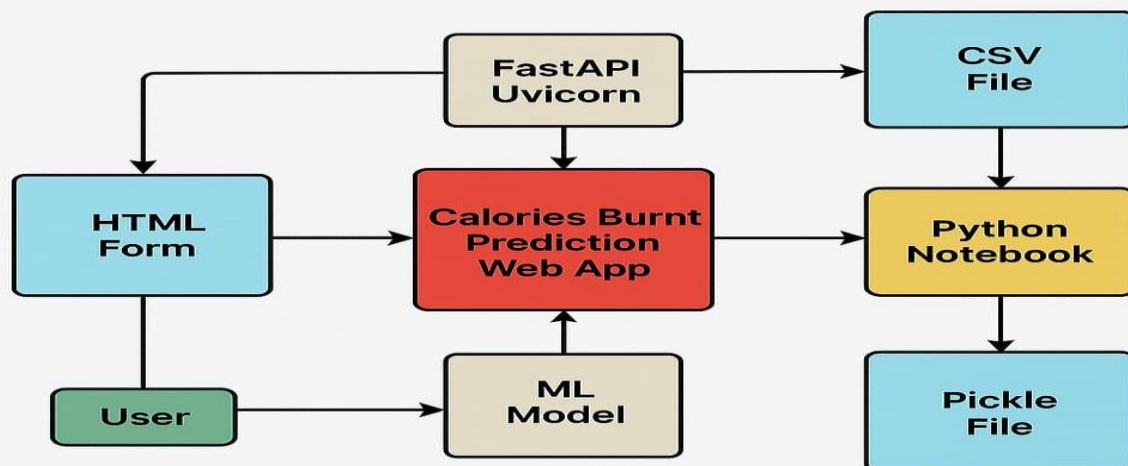
Symbol	Description
	Start/End: Represents the start or end point of a flowchart.
	Data Flow: Data Flow are pipelines through the packets of information flows.
	Decision: Decision decides the next step will be proceed or not. Y/N, T/F
	Input/Output: Represents input and output data.

Fig: DFD Symbol Diagram

DATA FLOW DIAGRAM



Level 0 DFD Diagram



Level 1 DFD Diagram

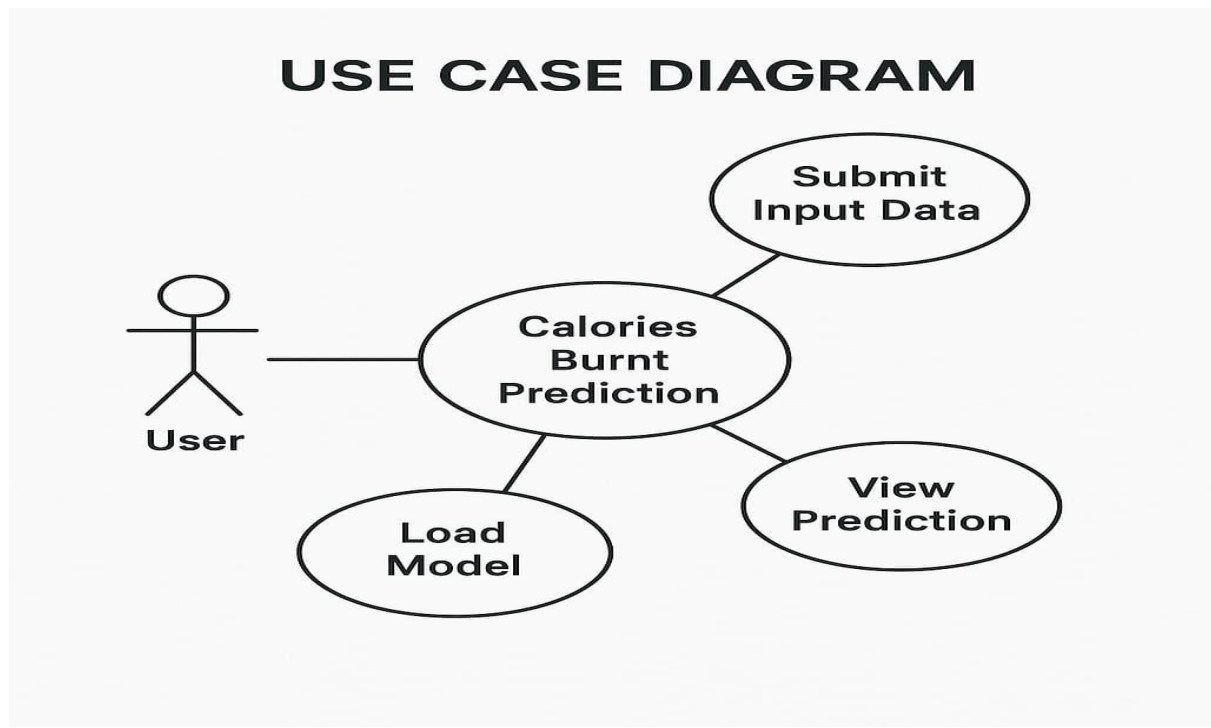
- **Use Case Description:**

Use Case Diagram:

- A **Use Case Diagram** is a type of **UML (Unified Modeling Language)** diagram that shows **how users (actors)** interact with a **system**. It focuses on **what the system does** — not **how** it does it.
- It's like a **visual map** of the features of your application from the user's point of view.

Actors:

- **User (Visitor):** interacts with the frontend
- **System (Backend/ML Model):** processes data and returns predictions
- **Admin/Data Scientist (optional):** uploads or retrains model



Design

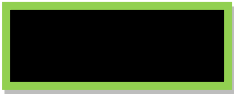
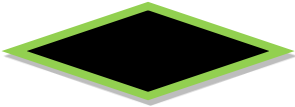
Design in software development is a broad but super important concept—it's all about planning *how* the software will be structured and function **before** (and during) the actual building. Think of it like the blueprint before constructing a building.

4.1 ER Diagram

4.2 Language and software used

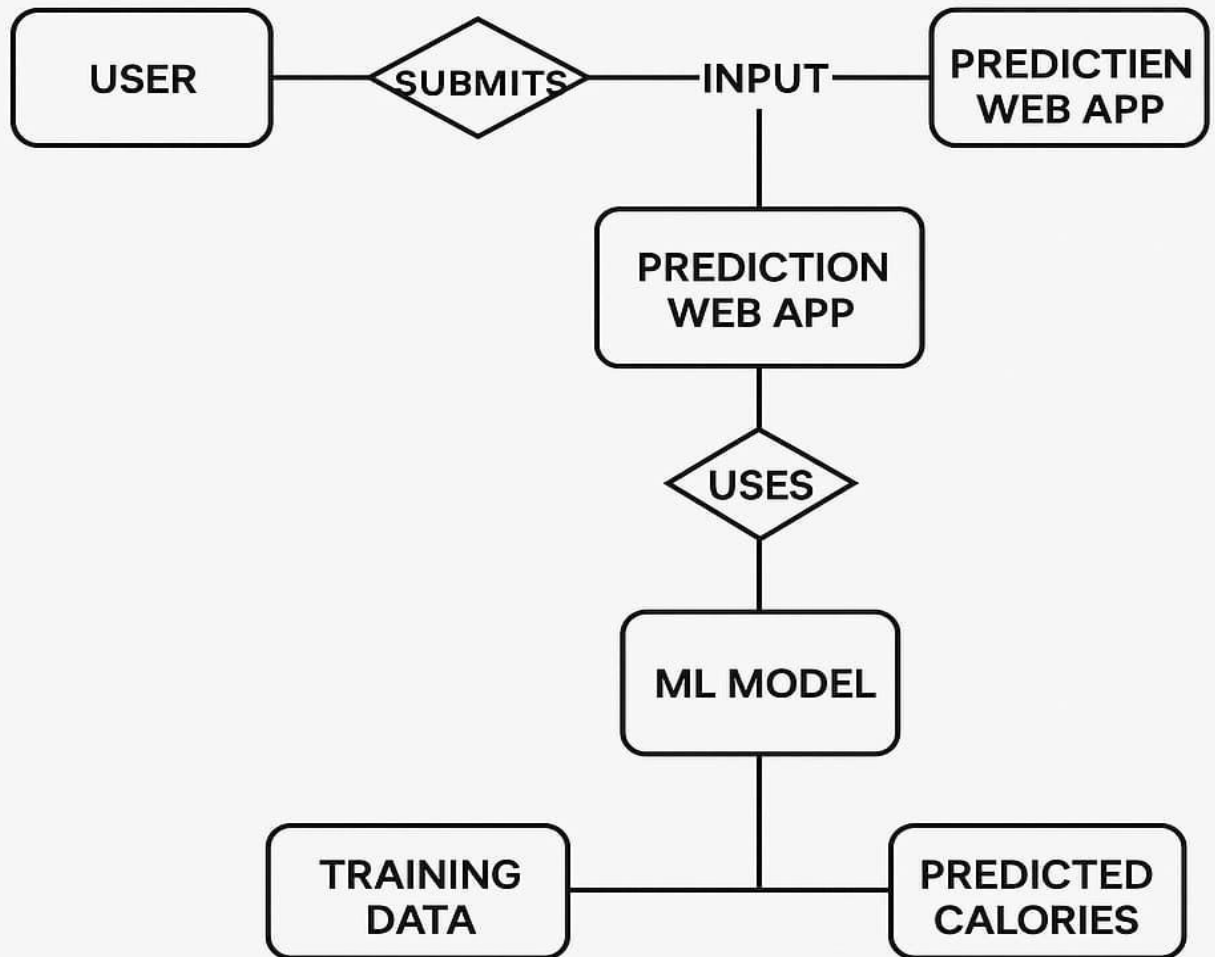
4.1 E-R Diagram

Entity – Relationship Diagram is a graphical representation of entities and their relationship to each other's – describes how data is related to each an entity is a piece of data – an object is a concept where data is stored, relationship how the data is shared entities.

Symbol	Name	Description
	Entity	Any entity can be any object which information is recorded.
	Attributes	An attribute represents the characteristics of an entity
	Relationship	A Relationship Describes relation between entities

ER Diagram Table

ER DIAGRAM



ER Diagram of Calories burnt Prediction

4.2 Language and software used

1. Python

- **Role in the Application:** Python is the primary language for both the **backend** and the **machine learning (ML) model**. The machine learning model is built using Python-based libraries, and the backend server is powered by **FastAPI**. Python's simple syntax and large collection of scientific libraries make it ideal for building ML-based applications.
- **Why Python:** Python is the de facto standard for data science and machine learning, making it the perfect choice for handling data, training models, and serving predictions via a web interface.

2. FastAPI

- **Role in the Application:** FastAPI is a modern, high-performance web framework for building APIs with Python. It handles HTTP requests (such as GET and POST requests) and serves the **ML model** to users via the web interface.
- **Why FastAPI:** FastAPI is chosen because it's built for high performance and is especially efficient for serving machine learning models. It supports asynchronous programming and is designed to be easy to use, with automatic data validation (via Pydantic) and documentation generation (via OpenAPI).

3. Uvicorn

- **Role in the Application:** Uvicorn is an ASGI server that serves FastAPI applications. It runs the application, manages incoming HTTP requests, and ensures the app performs efficiently by handling concurrent connections.
- **Why Uvicorn:** Uvicorn is fast and lightweight. It's built for handling asynchronous applications like FastAPI, providing excellent performance and scalability.

4. Jinja2

- **Role in the Application:** Jinja2 is used to dynamically generate HTML content by injecting Python data into HTML templates. It allows you to **render user-facing web pages** and display dynamic content such as the results of the calories burnt prediction.
- **Why Jinja2:** Jinja2 is a robust templating engine that integrates seamlessly with FastAPI. It allows for a clean separation of backend logic (Python) and frontend (HTML), making the codebase easier to maintain and extend.

5. Pickle

- **Role in the Application:** Pickle is used to **serialize** (save) and **deserialize** (load) the trained machine learning model and its preprocessing pipeline. After training the model in the Jupyter notebook, it's saved to disk and later loaded by the FastAPI app for real-time predictions.
- **Why Pickle:** Pickle is Python's built-in solution for serializing Python objects. It allows you to save complex objects (like machine learning models) to a file and later load them into memory when needed, making it ideal for serving pre-trained models in a web application.

6. HTML (Hypertext Markup Language)

- **Role in the Application:** HTML is the standard language for creating the structure and content of web pages. The app uses HTML (with Jinja2 templating) to define the user interface (UI), including input forms for the user to enter data and displays for showing prediction results.
- **Why HTML:** HTML provides the backbone for building the frontend of the web application. With Jinja2 rendering, HTML can display dynamic content, such as the predicted number of calories burnt.

7. CSV (Comma-Separated Values)

- **Role in the Application:** The CSV file stores the **sample data** used for training the machine learning model. It contains columns such as age, weight, height, and calories burnt, which are used to train the model.
- **Why CSV:** CSV is a simple, human-readable format that is easy to load and manipulate using Python's **pandas** library. It's a common choice for working with datasets in machine learning, particularly when the dataset is not very large.

Machine Learning Libraries and Tools:

8. Jupyter Notebook

- **Role in the Application:** Jupyter notebooks are used for **exploratory data analysis (EDA)**, **data preprocessing**, and **model training**. The calories burnt prediction model is trained in a notebook before it is serialized and used in the FastAPI app.
- **Why Jupyter Notebook:** Jupyter notebooks are an interactive environment that is excellent for experimentation, visualizations, and iterative coding. They allow you to combine code, visualizations, and markdown documentation in a single document, making it easy to track model development.

9. NumPy

- **Role in the Application:** NumPy is used for **numerical operations**, particularly for manipulating arrays and matrices. It is essential for handling large amounts of data and performing mathematical operations in the machine learning pipeline.
- **Why NumPy:** NumPy is the core scientific computing library in Python. It is used throughout the machine learning process for tasks like matrix operations, data transformations, and feature engineering.

10.Pandas

- **Role in the Application:** Pandas is used for **data manipulation and preprocessing**. It's used to load the CSV file, clean the data, handle missing values, and transform the data before passing it to the machine learning model.
- **Why Pandas:** Pandas is the go-to library for working with tabular data. It makes it easy to load, filter, aggregate, and manipulate data, which is essential for preparing the data before feeding it into the model.

11.Matplotlib & Seaborn

- **Role in the Application:** These libraries are used for **data visualization**. They are used to create plots and charts for **exploratory data analysis (EDA)**, such as histograms, scatter plots, and correlations.
- **Why Matplotlib & Seaborn:** These are the standard libraries for creating static, animated, and interactive visualizations in Python. Seaborn builds on Matplotlib and provides a high-level interface for creating attractive and informative statistical graphics.

12.scikit-learn (sklearn)

- **Role in the Application:** scikit-learn is used for **preprocessing, model training, and model evaluation**. It includes tools for scaling, splitting data, and evaluating the performance of machine learning models.
- **Why scikit-learn:** scikit-learn is a comprehensive machine learning library that provides simple and efficient tools for data mining and analysis. It's used for tasks like scaling features, training models, and cross-validation.

13.XGBoost

- **Role in the Application:** XGBoost is the **machine learning algorithm** used for predicting the calories burnt. It's an implementation of **gradient boosting**, known for its high performance and scalability, especially with large datasets.
- **Why XGBoost:** XGBoost is a popular algorithm for structured/tabular data. It's highly effective in predictive modeling tasks, including regression and classification, due to its robustness and ability to handle complex patterns in the data.

14.sklearn.pipeline

- **Role in the Application:** The pipeline is used to **chain multiple steps** (like preprocessing and model training) into a single workflow. This ensures that all preprocessing steps are applied consistently during both training and prediction.
- **Why Pipeline:** Using a pipeline ensures that data transformations (like scaling) are applied correctly during training and prediction, without leaking any test data into the training process. It also streamlines the code, making it more readable and reusable.

Software Used:

1. Visual Studio Code (VS Code)

- **Role in the Application:** VS Code is the **IDE (Integrated Development Environment)** used to write, edit, and manage the Python code, HTML templates, and other files in the project. It provides features like **syntax highlighting**, **auto-completion**, **linting**, and **integrated terminal** for running the FastAPI app and Jupyter notebooks.
- **Why VS Code:** VS Code is a lightweight, yet powerful IDE that supports a variety of programming languages and frameworks, making it ideal for full-stack development. It also has excellent support for Python, FastAPI, and Jupyter notebooks, along with extensions for debugging and version control.

2. Jupyter Notebook

- **Role in the Application:** Used for **model development** and data exploration. The model is developed, trained, and saved in Jupyter notebooks, before being integrated into the web app.
- **Why Jupyter:** It's an interactive environment that allows for easy experimentation and documentation. The ability to write code and view outputs immediately makes it perfect for machine learning development.

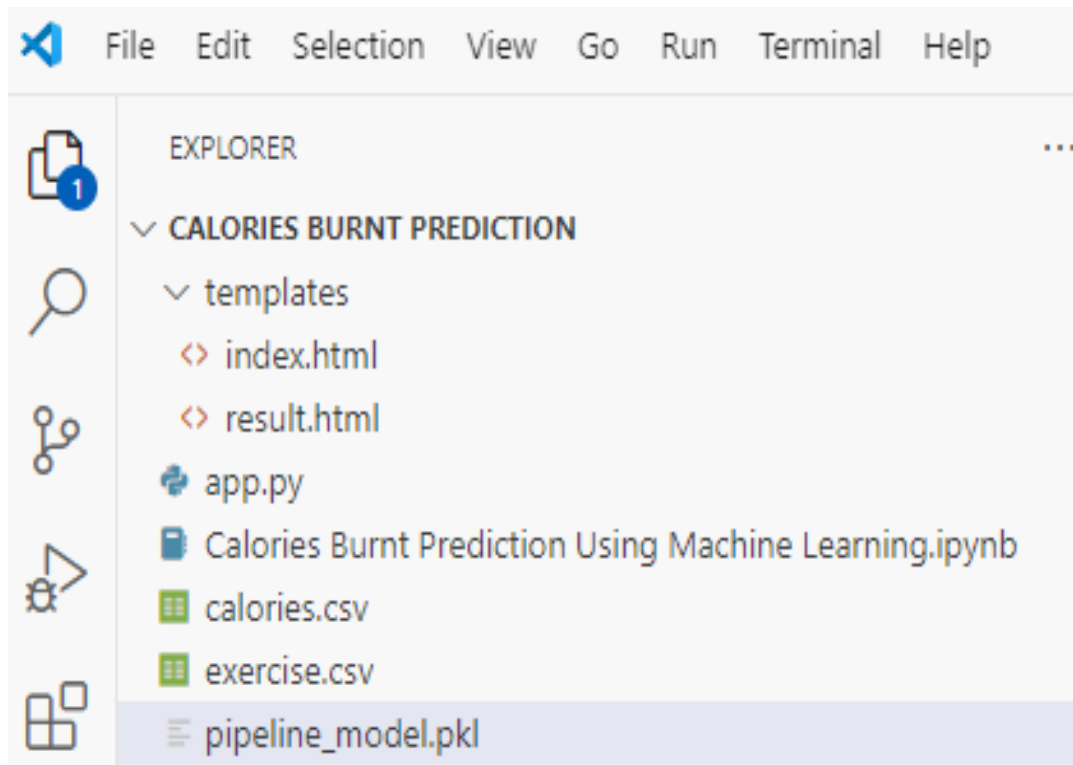
Deployment/Execution Environment:

- **Local Development:** The web application is developed and run locally in a development environment using **FastAPI** and **Uvicorn**. VS Code is used for coding, and **Uvicorn** serves the FastAPI app on your local machine.

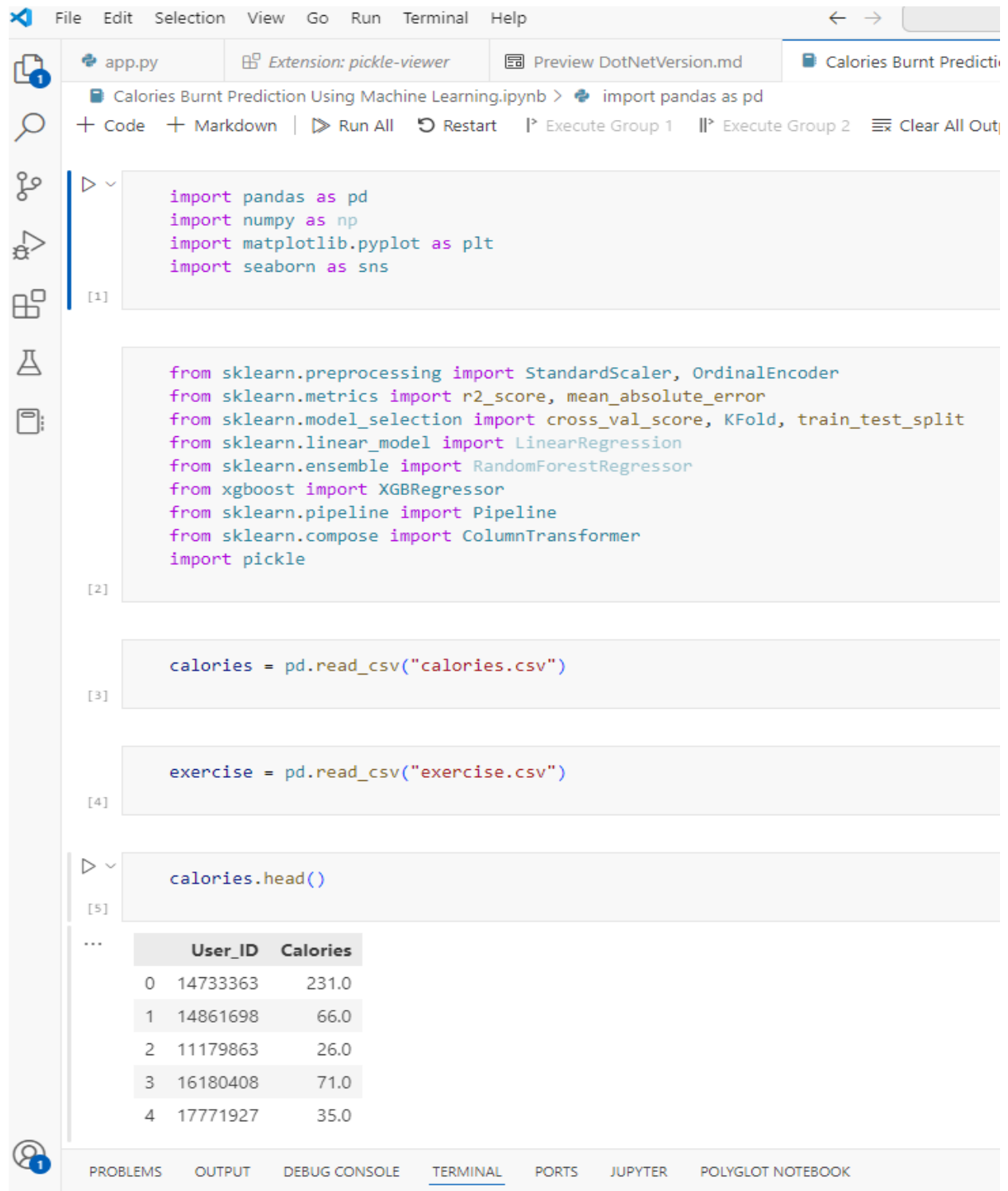
Summary:

- **Languages:** Python (backend & ML model), HTML (frontend).
- **Frameworks:** FastAPI (API), Uvicorn (ASGI server), Jinja2 (templating).
- **Libraries:** NumPy, Pandas, Matplotlib, Seaborn, scikit-learn, XGBoost, Pickle.
- **Data:** CSV (input data).
- **Software used :** Visual Studio Code (VS Code)

CODING



CALORIES BURNT PREDICTION (COLAB NOTEBOOK) IN VSCODE



The screenshot shows the Visual Studio Code (VS Code) interface with a Jupyter Notebook open. The notebook is titled "Calories Burnt Prediction Using Machine Learning.ipynb". The interface includes a top menu bar (File, Edit, Selection, View, Go, Run, Terminal, Help) and a sidebar on the left with icons for Explorer, Search, Source Control, Run and Debug, Extensions, Testing, and Data Science. The main editor area displays the notebook cells, which are numbered [1] through [5].

Cell [1] contains the following import statements:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Cell [2] contains the following imports for machine learning models and preprocessing:

```
from sklearn.preprocessing import StandardScaler, OrdinalEncoder
from sklearn.metrics import r2_score, mean_absolute_error
from sklearn.model_selection import cross_val_score, KFold, train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.ensemble import RandomForestRegressor
from xgboost import XGBRegressor
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
import pickle
```

Cell [3] contains the following code to load the "calories.csv" file:

```
calories = pd.read_csv("calories.csv")
```

Cell [4] contains the following code to load the "exercise.csv" file:

```
exercise = pd.read_csv("exercise.csv")
```

Cell [5] contains the following code to display the first few rows of the "calories" dataset:

```
calories.head()
```

The output of Cell [5] is displayed as a table with 5 rows and 2 columns: "User_ID" and "Calories".

	User_ID	Calories
0	14733363	231.0
1	14861698	66.0
2	11179863	26.0
3	16180408	71.0
4	17771927	35.0

The bottom status bar of VS Code shows the following tabs: PROBLEMS, OUTPUT, DEBUG CONSOLE, **TERMINAL**, PORTS, JUPYTER, and POLYGLOT NOTEBOOK.

Calories Burnt Prediction Using Machine Learning.ipynb > import pandas as pd

Code + Markdown Run All Restart Execute Group 1 Execute Group 2 Clear All Outputs

```
exercise.head()
```

[6]

```
...
```

	User_ID	Gender	Age	Height	Weight	Duration	Heart_Rate	Body_Temp
0	14733363	female	68	190	94	29	105	40.8
1	14861698	female	20	166	60	14	94	40.3
2	11179863	male	69	179	79	5	88	38.7
3	16180408	female	34	179	71	13	100	40.5
4	17771927	female	27	154	58	10	81	39.8

```
data = pd.merge(calories, exercise, on = 'User_ID')
```

[7]

```
data.head()
```

[8]

```
...
```

	User_ID	Calories	Gender	Age	Height	Weight	Duration	Heart_Rate	Body_Temp
0	14733363	231.0	female	68	190	94	29	105	40.8
1	14861698	66.0	female	20	166	60	14	94	40.3
2	11179863	26.0	male	69	179	79	5	88	38.7
3	16180408	71.0	female	34	179	71	13	100	40.5
4	17771927	35.0	female	27	154	58	10	81	39.8

>

```
data.shape
```

[9]

```
...
```

(15000, 9)

```
data.info()
```

[10]

app.py

Extension: pickle-viewer

Preview DotNetVersion.md

Calories Burnt Prediction Using Machine Learning.ipynb

Pre

Calories Burnt Prediction Using Machine Learning.ipynb > import pandas as pd

+ Code + Markdown | ▶ Run All ↺ Restart | ▶ Execute Group 1 || ▶ Execute Group 2 | ⌵ Clear All Outputs | 📄 Jupyter Variables | 📖 Outline ..

▶ data.info()

[10]

...<class 'pandas.core.frame.DataFrame'>
RangeIndex: 15000 entries, 0 to 14999
Data columns (total 9 columns):
Column Non-Null Count Dtype

0 User_ID 15000 non-null int64
1 Calories 15000 non-null float64
2 Gender 15000 non-null object
3 Age 15000 non-null int64
4 Height 15000 non-null int64
5 Weight 15000 non-null int64
6 Duration 15000 non-null int64
7 Heart_Rate 15000 non-null int64
8 Body_Temp 15000 non-null float64
dtypes: float64(2), int64(6), object(1)
memory usage: 1.0+ MB

data.describe()

[11]

...

	User_ID	Calories	Age	Height	Weight	Duration	Heart_Rate	Body_Temp
count	1.500000e+04	15000.000000	15000.000000	15000.000000	15000.000000	15000.000000	15000.000000	15000.000000
mean	1.497736e+07	89.539533	42.789800	174.465133	74.966867	15.530600	95.518533	40.025453
std	2.872851e+06	62.456978	16.980264	14.258114	15.035657	8.319203	9.583328	0.779230
min	1.000116e+07	1.000000	20.000000	123.000000	36.000000	1.000000	67.000000	37.100000
25%	1.247419e+07	35.000000	28.000000	164.000000	63.000000	8.000000	88.000000	39.600000
50%	1.499728e+07	79.000000	39.000000	175.000000	74.000000	16.000000	96.000000	40.200000
75%	1.744928e+07	138.000000	56.000000	185.000000	87.000000	23.000000	103.000000	40.600000
max	1.999965e+07	314.000000	79.000000	222.000000	132.000000	30.000000	128.000000	41.500000

Code + Markdown | ▶ Run All ↺ Rest

data.isnull().sum()

[12]

...User_ID 0
Calories 0
Gender 0
Age 0
Height 0
Weight 0
Duration 0
Heart_Rate 0
Body_Temp 0
dtype: int64

Code + Markdown | ▶ Run All ↺ Restart | ▶ Execute Group 1 || ▶ Execute Group 2 ☰ Clear All Outputs |

```
data.duplicated().any()
```

1

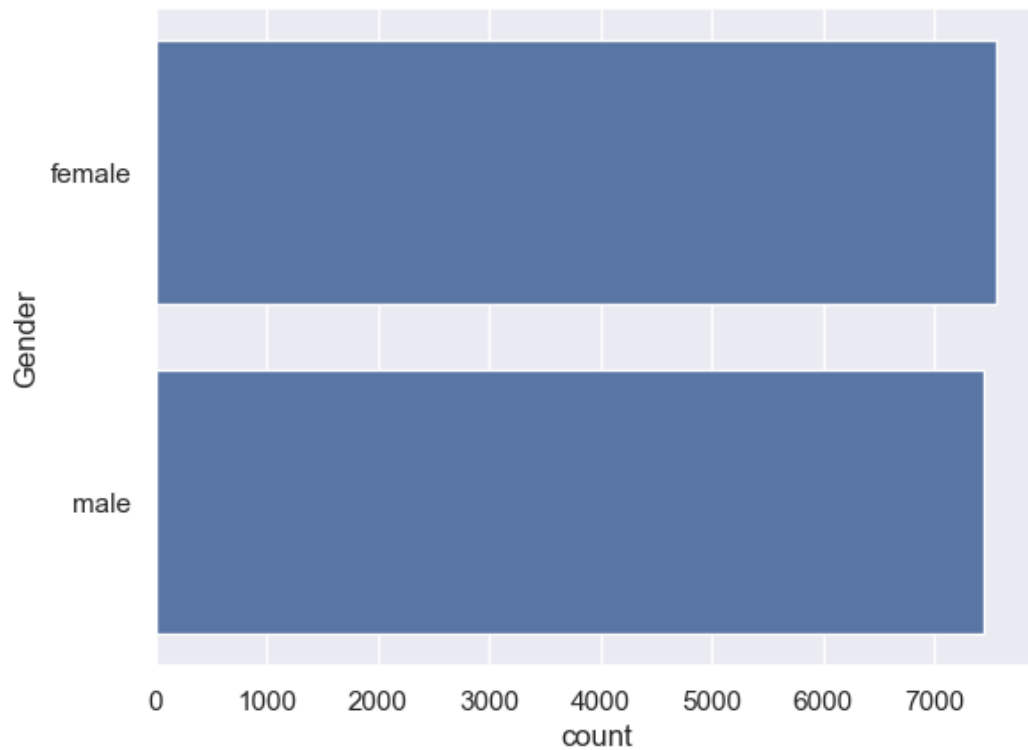
np.False_

Data Visualization

```
sns.set()  
sns.countplot(data['Gender'])
```

1

<Axes: xlabel='count', ylabel='Gender'>



4

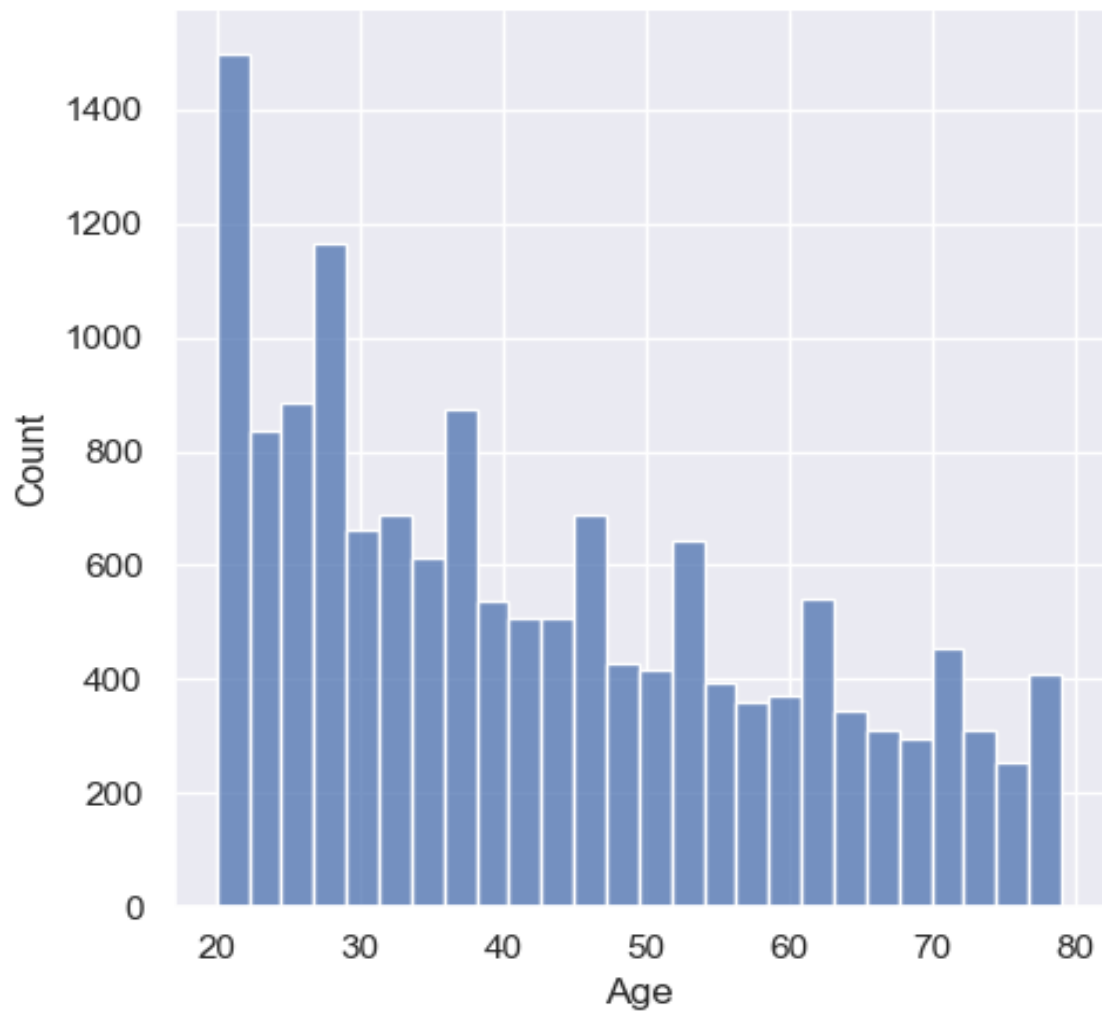
> ~

```
sns.displot(data['Age'])
```

16]

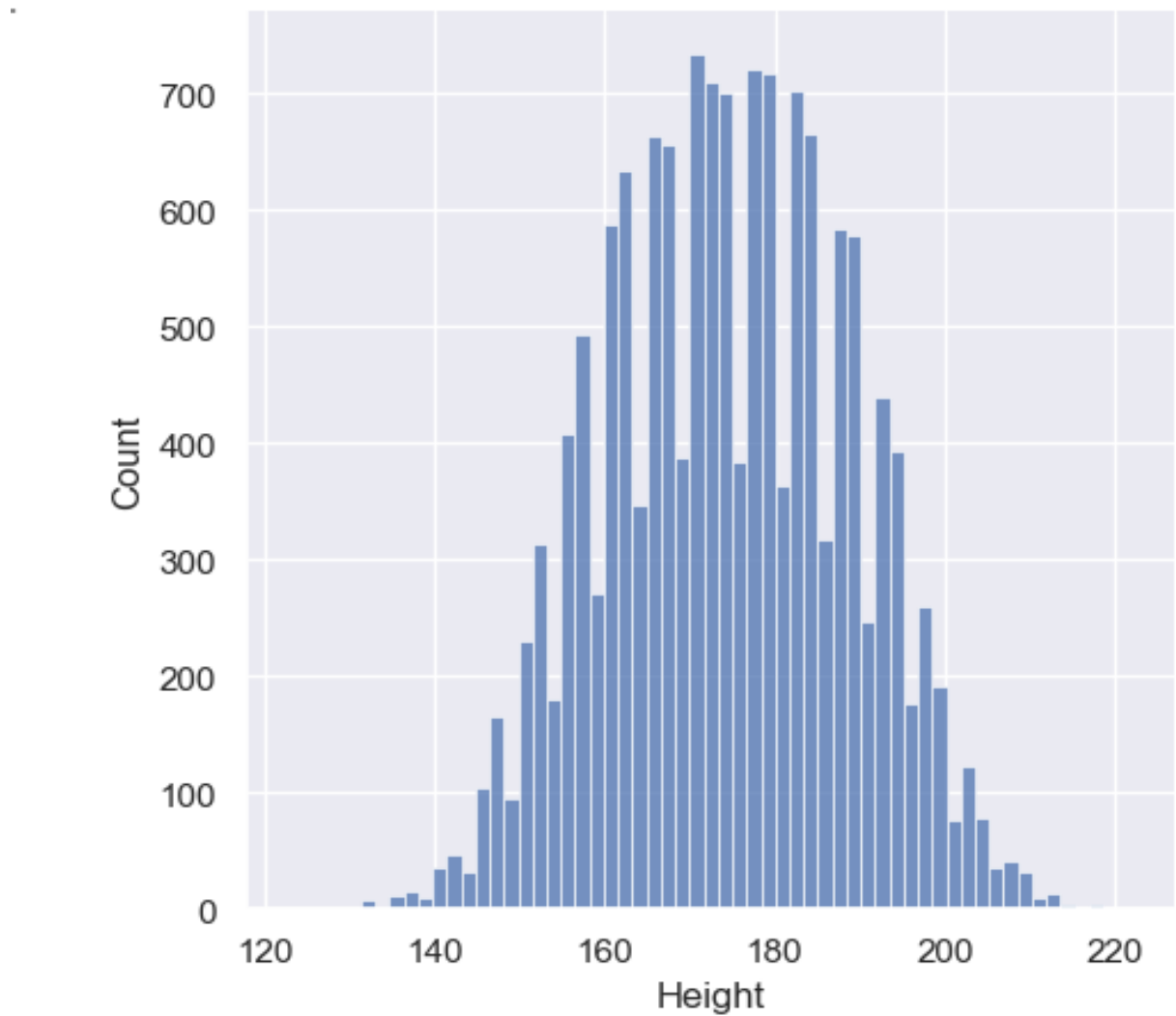
.. <seaborn.axisgrid.FacetGrid at 0x13946987ef0>

..



```
7] sns.displot(data['Height'])
```

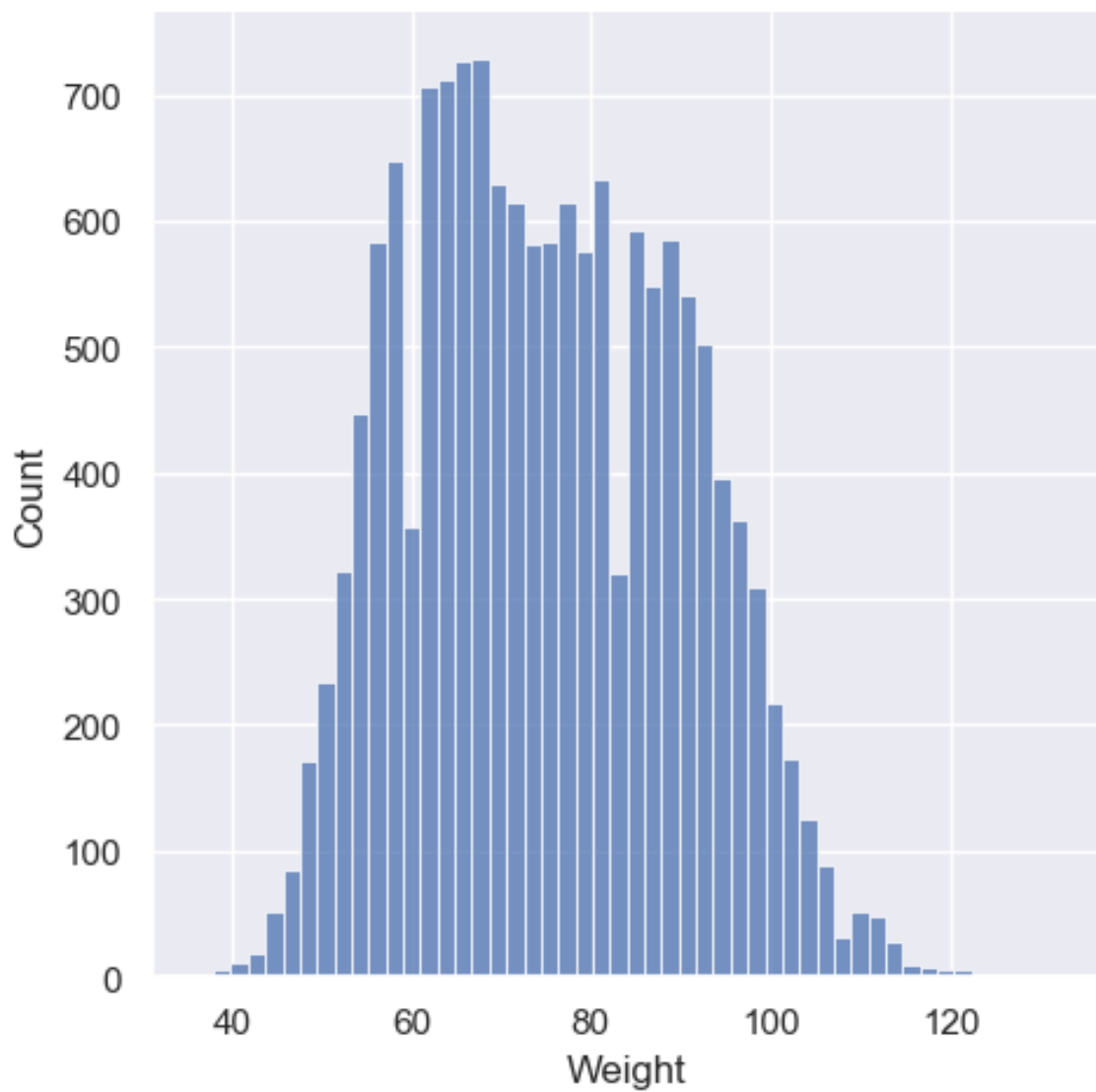
• <seaborn.axisgrid.FacetGrid at 0x1394842fd70>



✓
]

```
sns.displot(data['Weight'])
```

<seaborn.axisgrid.FacetGrid at 0x1394a644080>



◀

0]

```
data['Gender']=data['Gender'].map({'male':0,'female':1})
```

1]

```
data.corr()
```

	User_ID	Calories	Gender	Age	Height	Weight	Duration	Heart_Rate	Body_Temp
User_ID	1.000000	-0.001661	0.000675	-0.001827	-0.013520	-0.011603	-0.002751	-0.000457	0.000923
Calories	-0.001661	1.000000	-0.022055	0.154395	0.017537	0.035481	0.955421	0.897882	0.824558
Gender	0.000675	-0.022055	1.000000	-0.003024	-0.710390	-0.783018	-0.003224	-0.011423	-0.007131
Age	-0.001827	0.154395	-0.003024	1.000000	0.009554	0.090094	0.013247	0.010482	0.013175
Height	-0.013520	0.017537	-0.710390	0.009554	1.000000	0.958451	-0.004625	0.000528	0.001200
Weight	-0.011603	0.035481	-0.783018	0.090094	0.958451	1.000000	-0.001884	0.004311	0.004095
Duration	-0.002751	0.955421	-0.003224	0.013247	-0.004625	-0.001884	1.000000	0.852869	0.903167
Heart_Rate	-0.000457	0.897882	-0.011423	0.010482	0.000528	0.004311	0.852869	1.000000	0.771529
Body_Temp	0.000923	0.824558	-0.007131	0.013175	0.001200	0.004095	0.903167	0.771529	1.000000

+ Code

+ Markdown

2]

```
correlation = data.corr()
```

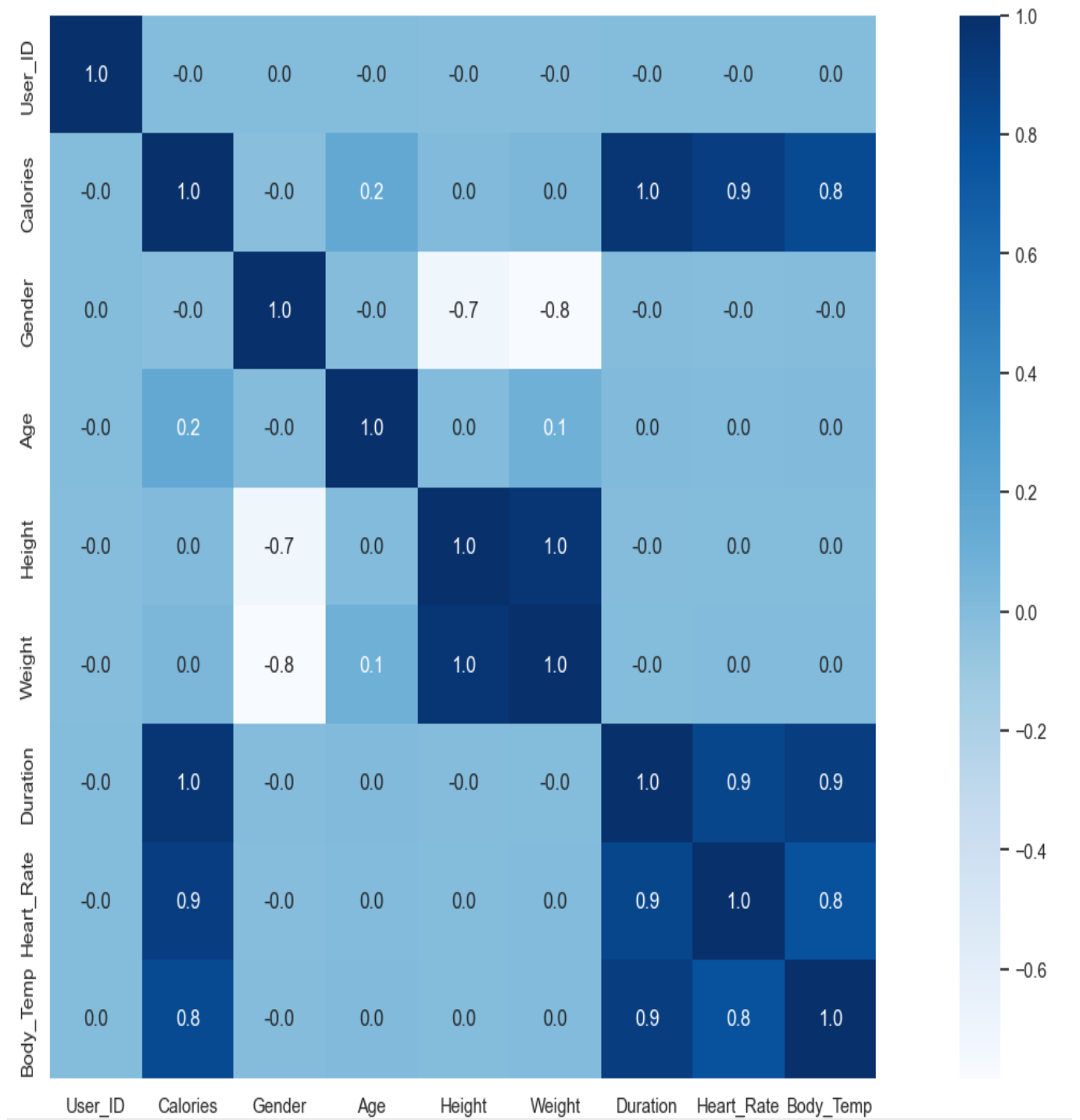
3]

```
correlation
```

	User_ID	Calories	Gender	Age	Height	Weight	Duration	Heart_Rate	Body_Temp
User_ID	1.000000	-0.001661	0.000675	-0.001827	-0.013520	-0.011603	-0.002751	-0.000457	0.000923
Calories	-0.001661	1.000000	-0.022055	0.154395	0.017537	0.035481	0.955421	0.897882	0.824558
Gender	0.000675	-0.022055	1.000000	-0.003024	-0.710390	-0.783018	-0.003224	-0.011423	-0.007131
Age	-0.001827	0.154395	-0.003024	1.000000	0.009554	0.090094	0.013247	0.010482	0.013175
Height	-0.013520	0.017537	-0.710390	0.009554	1.000000	0.958451	-0.004625	0.000528	0.001200
Weight	-0.011603	0.035481	-0.783018	0.090094	0.958451	1.000000	-0.001884	0.004311	0.004095
Duration	-0.002751	0.955421	-0.003224	0.013247	-0.004625	-0.001884	1.000000	0.852869	0.903167
Heart_Rate	-0.000457	0.897882	-0.011423	0.010482	0.000528	0.004311	0.852869	1.000000	0.771529
Body_Temp	0.000923	0.824558	-0.007131	0.013175	0.001200	0.004095	0.903167	0.771529	1.000000

4]

```
plt.figure(figsize = (20, 10))
sns.heatmap(correlation, cbar = True, square=True, fmt = ".1f", annot=True, cmap = 'Blues')
```



```
X = data.drop(columns = ['User_ID', 'Calories'])
y = data['Calories']
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2, random_state = 42)
```

```
data.columns
```

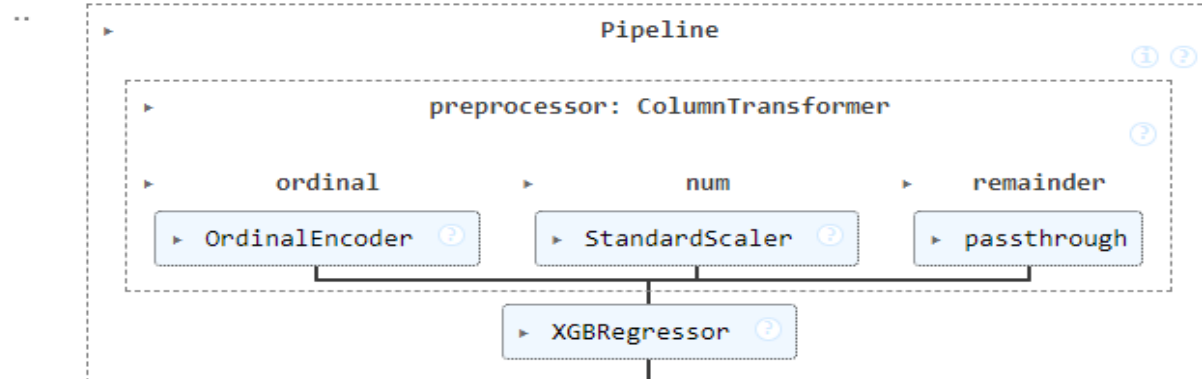
```
Index(['User_ID', 'Calories', 'Gender', 'Age', 'Height', 'Weight', 'Duration',
      'Heart_Rate', 'Body_Temp'],
      dtype='object')
```

```
preprocessor = ColumnTransformer(transformers=[
    ('ordinal', OrdinalEncoder(), ['Gender']),
    ('num', StandardScaler(), ['Age', 'Height', 'Weight', 'Duration', 'Heart_Rate', 'Body_Temp'])
], remainder='passthrough')
```

```
pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('model', XGBRegressor())
])
```

```
> ✓ pipeline.fit(X_train, y_train)
```

```
30]
```



```
y_pred = pipeline.predict(X_test)
```

```
31]
```

```
r2_score(y_test, y_pred)
```

```
32]
```

```
.. 0.9988255884401388
```

```
mean_absolute_error(y_test, y_pred)
```

```
33]
```

```
.. 1.4955601506034533
```

```
kfold = KFold(n_splits=5, shuffle=True, random_state=42)  
cv_results = cross_val_score(pipeline, X, y, cv = kfold, scoring = 'r2')
```

```
34]
```

```
cv_results.mean()
```

```
1
```

```
np.float64(0.9987992870477731)
```

```
with open('pipeline_model.pkl', 'wb') as f:  
    pickle.dump(pipeline, f)
```

```
1
```

PYTHON CODE

```
app.py  ×  Extension: pickle-viewer  Preview DotNetVersion.md  Calories Burnt Prediction Using Machine Le

app.py > predict > Body_Temp
1  from fastapi import FastAPI, Form, Request, Depends
2  from fastapi.templating import Jinja2Templates
3  import pickle
4  import pandas as pd
5
6
7  app = FastAPI()
8
9  # Load the ML model
10 with open("pipeline_model.pkl", "rb") as f:
11     pipeline = pickle.load(f)
12
13 # Setup templates directory
14 templates = Jinja2Templates(directory="templates")
15
16
17 @app.get("/")
18 def home(request: Request):
19     return templates.TemplateResponse("index.html", {"request": request})
20
21 @app.post("/predict")
22 def predict(
23     request: Request,
24     Gender: str = Form(...),
25     Age: float = Form(...),
26     Height: float = Form(...),
27     Weight: float = Form(...),
28     Duration: float = Form(...),
29     Heart_Rate: float = Form(...),
30     Body_Temp: float = Form(...)
31 ):
32     sample = pd.DataFrame({
33         "Gender": [Gender],
34         "Age": [Age],
35         "Height": [Height],
36         "Weight": [Weight],
37         "Duration": [Duration],
38         "Heart_Rate": [Heart_Rate],
39         "Body_Temp": [Body_Temp]
40     }, index=[0])
41
42     result = pipeline.predict(sample)[0]
43
44     return templates.TemplateResponse("result.html", {"request": request, "calories": result})
45
46 if __name__ == "__main__":
47     import uvicorn
48     uvicorn.run(app, port=8000)
```

INDEX.HTML FILE

```
ip.py  index.html x Preview DotNetVersion.md Calories Burnt Prediction Using Machine Learning.ipynb Preview DotNetVersion.md Preview DotNetVersion.md calories.csv

plates > index.html > ...
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
5   <title>Calories Burnt Prediction</title>
7   <script src="https://cdn.tailwindcss.com"></script>
3 </head>
3 <body class="bg-gray-100 flex items-center justify-center min-h-screen">
3   <div class="bg-white p-8 rounded-2xl shadow-lg w-full max-w-md">
1     <h2 class="text-2xl font-bold text-center text-gray-800 mb-6">Calories Burnt Prediction</h2>
2     <form action="/predict" method="post" class="space-y-4">
3       <div>
4         <label class="block text-gray-700 font-medium">Gender</label>
5         <select name="Gender" class="w-full px-4 py-2 mt-1 border rounded-lg focus:ring-2 focus:ring-blue-400 outline-none">
5           <option value="male">Male</option>
7           <option value="female">Female</option>
3         </select>
3       </div>
3       <div>
1         <label class="block text-gray-700 font-medium">Age</label>
2         <input type="number" step="1" name="Age" class="w-full px-4 py-2 mt-1 border rounded-lg focus:ring-2 focus:ring-blue-400 outline-none" required>
3       </div>
4       <div>
5         <label class="block text-gray-700 font-medium">Height (cm)</label>
5         <input type="number" step="0.1" name="Height" class="w-full px-4 py-2 mt-1 border rounded-lg focus:ring-2 focus:ring-blue-400 outline-none" required>
7       </div>
3       <div>
3         <label class="block text-gray-700 font-medium">Weight (kg)</label>
3         <input type="number" step="0.1" name="Weight" class="w-full px-4 py-2 mt-1 border rounded-lg focus:ring-2 focus:ring-blue-400 outline-none" required>
1       </div>
2       <div>
3         <label class="block text-gray-700 font-medium">Duration (min)</label>
4         <input type="number" step="0.1" name="Duration" class="w-full px-4 py-2 mt-1 border rounded-lg focus:ring-2 focus:ring-blue-400 outline-none" required>
5       </div>
5       <div>
7         <label class="block text-gray-700 font-medium">Heart Rate</label>
3         <input type="number" step="0.1" name="Heart_Rate" class="w-full px-4 py-2 mt-1 border rounded-lg focus:ring-2 focus:ring-blue-400 outline-none" required>
3       </div>
3       <div>
1         <label class="block text-gray-700 font-medium">Body Temperature (°C)</label>
2         <input type="number" step="0.1" name="Body_Temp" class="w-full px-4 py-2 mt-1 border rounded-lg focus:ring-2 focus:ring-blue-400 outline-none" required>
3       </div>
4       <button type="submit" class="w-full bg-blue-600 hover:bg-blue-700 text-white font-medium py-2 rounded-lg shadow-md transition duration-300">Predict</button>
5     </form>
46   </div>
47 </body>
48 </html>
49
```

RESULT.HTML FILE

templates > <> result.html > ...

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8">
5   <meta name="viewport" content="width=device-width, initial-scale=1.0">
6   <title>Prediction Result</title>
7   <script src="https://cdn.tailwindcss.com"></script>
8 </head>
9 <body class="bg-gray-100 flex items-center justify-center min-h-screen">
10   <div class="bg-white p-8 rounded-2xl shadow-lg text-center w-full max-w-md">
11     <h2 class="text-2xl font-bold text-gray-800 mb-6">Prediction Result</h2>
12     <div class="bg-green-100 text-green-800 px-6 py-4 rounded-lg shadow-md">
13       <h4 class="text-lg font-semibold">Calories Burnt: <strong>{{ calories }}</strong> kcal</h4>
14     </div>
15     <div class="mt-6">
16       <a href="/" class="bg-blue-600 hover:bg-blue-700 text-white font-medium py-2 px-6 rounded-lg shadow-md transition duration-300">Go Back</a>
17     </div>
18   </div>
19 </body>
20 </html>
21
```

CALORIES AND EXERCISE CSV FILE DATA (UPTO 1500 RANDOM DATA)

calories.csv > data

1	User_ID,Calories
2	14733363,231.0
3	14861698,66.0
4	11179863,26.0
5	16180408,71.0
6	17771927,35.0
7	15130815,123.0
8	19602372,112.0
9	11117088,143.0
10	12132339,134.0
11	17964668,72.0
12	13723164,3.0
13	13681290,92.0
14	15566424,58.0
15	12891699,88.0
16	13823829,7.0
17	17557348,170.0
18	12198133,11.0
19	15236104,43.0
20	11042324,15.0
21	16864285,74.0
22	11674347,29.0
23	19797300,32.0
24	14711095,10.0
25	14434854,155.0
26	14893804,3.0
27	17231597,24.0
28	10901446,12.0
29	15874362,10.0
30	15569252,195.0
31	15615743,5.0
32	13363046,69.0
33	17572853,136.0
34	17157339,80.0
35	18328111,107.0
36	19303479,200.0
37	10699201,56.0
38	15283313,50.0
39	16324247,100.0
40	14277710,200.0
41	10888188,4.0
42	13379795,70.0
43	17181524,127.0
44	15988442,197.0
45	19538533,113.0

exercise.csv > data

	User_ID	Gender	Age	Height	Weight	Duration	Heart_Rate	Body_Temp
1	14733363	female	68	190	94	29	105	40.8
2	14861698	female	20	166	60	14	94	40.3
3	11179863	male	69	179	79	5	88	38.7
4	16180408	female	34	179	71	13	100	40.5
5	17771927	female	27	154	58	10	81	39.8
6	15130815	female	36	151	50	23	96	40.7
7	19602372	female	33	158	56	22	95	40.5
8	11117088	male	41	175	85	25	100	40.7
9	12132339	male	60	186	94	21	97	40.4
10	17964668	female	26	146	51	16	90	40.2
11	13723164	female	36	177	76	1	74	37.8
12	13681290	female	21	157	56	17	100	40
13	15566424	male	66	171	79	11	90	40
14	12891699	female	32	157	54	18	93	40.4
15	13823829	male	53	182	85	2	82	38.1
16	17557348	female	39	156	62	28	104	40.8
17	12198133	male	39	182	82	4	82	38.6
18	15236104	male	46	169	67	11	89	40.2
19	11042324	female	27	171	65	4	85	38.6
20	16864285	male	50	188	86	14	94	40.2
21	11674347	male	67	189	93	8	77	39.2
22	19797300	female	31	148	50	8	84	39.5
23	14711095	female	33	157	60	3	80	38.7
24	14434854	female	20	165	59	29	100	41
25	14893804	male	48	182	85	1	80	37.7
26	17231597	male	29	176	75	10	83	39.7
27	10901446	male	33	173	73	7	78	39.3
28	15874362	male	42	190	88	3	83	38.9
29	15569252	female	62	159	59	29	106	41.2
30	15615743	male	38	171	75	2	81	38.2
31	13363046	male	20	183	88	16	97	40.5
32	17572853	female	25	160	59	24	102	40.3
33	17157339	female	24	165	59	18	91	40.2
34	18328111	female	42	165	68	22	93	40.8
35	19303479	male	22	182	84	29	114	41
36	10699201	female	74	158	59	10	93	39.6
37	15283313	female	70	154	59	10	88	40
38	16324247	female	26	182	80	21	96	40.5
39	14277710	male	44	184	86	25	114	40.8
40	10888188	male	61	183	86	1	81	38.3
41	13379795	female	68	157	57	13	92	40.1
42	17181524	female	61	176	70	20	104	40.5
43	15988442	male	63	179	80	25	108	40.8
44	19538533	female	54	171	66	20	98	40.1

Input/output Screen:

```
INFO: Started server process [5088]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: 127.0.0.1:55585 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:63983 - "GET / HTTP/1.1" 200 OK
```

```
INFO: Started server process [5088]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: 127.0.0.1:55585 - "GET / HTTP/1.1" 200 OK
INFO: 127.0.0.1:63983 - "GET / HTTP/1.1" 200 OK
```

Calories Burnt Prediction

×

+

127.0.0.1:8000

☆

Calories Burnt Prediction

Gender

Male

▼

Age

Height (cm)

Weight (kg)

Duration (min)

Heart Rate

Body Temperature (°C)

Predict

Calories Burnt Prediction

Gender

Female

Age

22

Height (cm)

163

Weight (kg)

80

Duration (min)

30

Heart Rate

84

Body Temperature (°C)

98

Predict

Prediction Result

Calories Burnt: **121.90834** kcal

[Go Back](#)

CONCLUSION

The **Calories Burnt Prediction Project** successfully demonstrates how **machine learning** can be integrated with **modern web technologies** to deliver accurate and real-time calorie expenditure estimates based on individual data and activity levels.

Objective Recap

The core objective of this project was to develop a **predictive system** that estimates the number of calories burned by an individual based on personal physiological data (such as age, gender, height, weight) and activity metrics (like duration and type of physical activity). The system aims to assist individuals in tracking fitness goals, optimizing workout routines, and promoting healthier lifestyles.

Achievements & Outcomes

1. Accurate Prediction Model

Using **machine learning algorithms**, particularly **XGBoost** integrated into a **Sklearn pipeline**, the model achieved high accuracy in predicting calorie expenditure. The model was trained and validated using a labeled dataset and fine-tuned for performance.

2. End-to-End Web Application

The solution includes a full-stack **web application** developed with:

- **FastAPI** for building the backend RESTful API,
- **Jinja2 and HTML** for rendering dynamic user interfaces,
- **Uvicorn** for serving the application,
- **Pickle** for saving and loading the trained ML model.

3. Data Analysis & Visualization

Comprehensive data preprocessing and analysis were performed using **NumPy, Pandas, Matplotlib, and Seaborn** to understand data trends and correlations, which helped improve model accuracy and interpretability.

4. User-Friendly Interaction

The app supports both **manual user input** and **bulk prediction using CSV file uploads**, making it versatile for individuals and potential integration into fitness platforms.

5. Learning & Technical Contributions

- Implemented ML lifecycle: **data collection, exploration, model training, evaluation, and deployment**
- Applied techniques like **feature engineering, pipeline construction, and model serialization**
- Developed a **responsive and scalable API** using FastAPI
- Enhanced skills in **frontend-backend integration** using Jinja2 templates
- Worked with **real-world health and fitness data**

Limitations

- The model relies heavily on the quality and diversity of the training data; real-world performance may vary based on user habits not captured in the dataset.
- Does not yet account for dynamic parameters such as workout intensity level, or individual metabolic rate, which could enhance accuracy.

Future Enhancements

- **Integration with wearables** (e.g., Fitbit, Apple Watch) to fetch real-time physical activity data.
- **Mobile version** of the web app for on-the-go accessibility.
- Use of **deep learning models** or **time series analysis** for more dynamic prediction.
- 📄 **User authentication & history tracking** to allow users to monitor progress over time.

References

- <https://github.com>
- <https://youtube.com>
- www.google.com
- <https://scikit-learn.org>
- <https://numpy.org>
- <https://pandas.pydata.org>
- <https://matplotlib.org>
- <https://seaborn.pydata.org>
- <https://fastapi.tiangolo.com>
- <https://www.uvicorn.org>
- <https://jinja.palletsprojects.com>